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1. (Math Question) Edges in a Graph

What is the smallest number of edges that an undirected connected graph on N vertices can have?

Pick **ONE** option

- N
- 2*N
- N-1
- 2*N-1

Clear Selection

2. (Math Question) Arranging vowels and consonants

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ALL

2. (Math Question) Arranging vowels and consonants

How many ways are there to arrange the letters in the word "ELEPHANT" such that no two vowels are adjacent?

Pick **ONE** option

- 720
- None of these
- 2880
- 5040

7200

Clear Selection

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3. (Math Question) Modulo of sum of powers

https://www.hackerrank.com/test/561nl7igp3m/questions/e498og8lmq4

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3. (Math Question) Modulo of sum of powers

What is the value of $(5^3 + 7^2 + 3^4) mod 9$?

Pick **ONE** option

- ( ) 0
- ( ) 9
- ( ) 3 ✓
- ( ) 7

Clear Selection

4. (Math Question) Sum of Infinite terms

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4. (Math Question) Sum of Infinite terms

Evaluate (sum from i=0 to inf) (2/7)^i

Pick **ONE** option

- 7/2
- 5/7
- 2/7
- 7/5

Clear Selection

5. (Math Question) Sum of reciprocal of HP terms

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5. (Math Question) Sum of reciprocal of HP terms

The sum of the reciprocals of the first 40 terms of a harmonic progression (HP) is 1600, and the sum of reciprocals of the first 60 terms is 3300. What is the sum of the reciprocals of the first 100 terms?

Pick **ONE** option

- 9500
- 11200
- 12600
- 10000

Clear Selection

8500

6. (Math Question) Sum of natural numbers

Answered: 2 / 44

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6. (Math Question) Sum of natural numbers

What are the values of a, b, c which satisfy the condition $a^n + b^n = c^n$, where a, b, c, n are all natural numbers and $n > 2$?

Pick **ONE** option

- a=3,b=4,c=5
- a=20,b=99,c=101
- a=25,b=50,c=75
- None of these ✓

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Clear Selection

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7. (Prog. Concept Question) Scheduling Algorithm

Which scheduling algorithm minimizes waiting time the most?

Pick **ONE** option

- Round Robin
- First Come First Serve
- Shortest Remaining Time First
- Shortest Job Next

Clear Selection

8. (Prog. Concept Question) Operators

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8. (Prog. Concept Question) Operators

Choose the incorrect remarks out of the following:

Pick **ONE OR MORE** options

- [ ] C++ allows any operator to be overloaded.
- [ ] Some of the existing operators cannot be overloaded.
- [ ] Operator precedence cannot be changed.

Clear Selection

9. (Prog. Concept Question) Page Fault

What can happen when a Page fault occurs?

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9. (Prog. Concept Question) Page Fault

What can happen when a Page fault occurs?

Pick **ONE OR MORE** options

- [ ] The running program terminates giving an error.
- [ ] The program tries to find the missing page in TLB and page table.
- [ ] The control is given to Kernel and Kernel tries to get the page into the main memory.

Clear Selection

10. (Prog. Concept Question) Bitwise Operators

If n & (n - 1) == 0, what can you say about integer n?

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10. (Prog. Concept Question) Bitwise Operators

If n & (n - 1) == 0, what can you say about integer n?

Pick **ONE** option

- n is an even number.
- n is a power of 2.
- n is an odd number.
- n is not a power of 2.

Clear Selection

11. (Programming Fundamentals) Diophantine Equations

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11. (Programming Fundamentals) Diophantine Equations

For a linear Diophantine equation, ax + by = c. What is the value of x, if a solution exists? (Where a, b, c are not zeros and g = gcd(a,b))

Pick **ONE** option

- X = (c/g) * (b/g) mod (a/g)
- X = (c/g) * (a/g)^-1 mod (b/g) ✓
- X = (c/g) * (a/g) mod (b/g)
- X = (c/g)* (b/g)^-1 mod (a/g)

Clear Selection

12. (Programming Fundamentals) Flow Cut

hackerrank.com/test/561nl7jqp3m/questions/brmjacqnc6b

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Answered: 2 / 44

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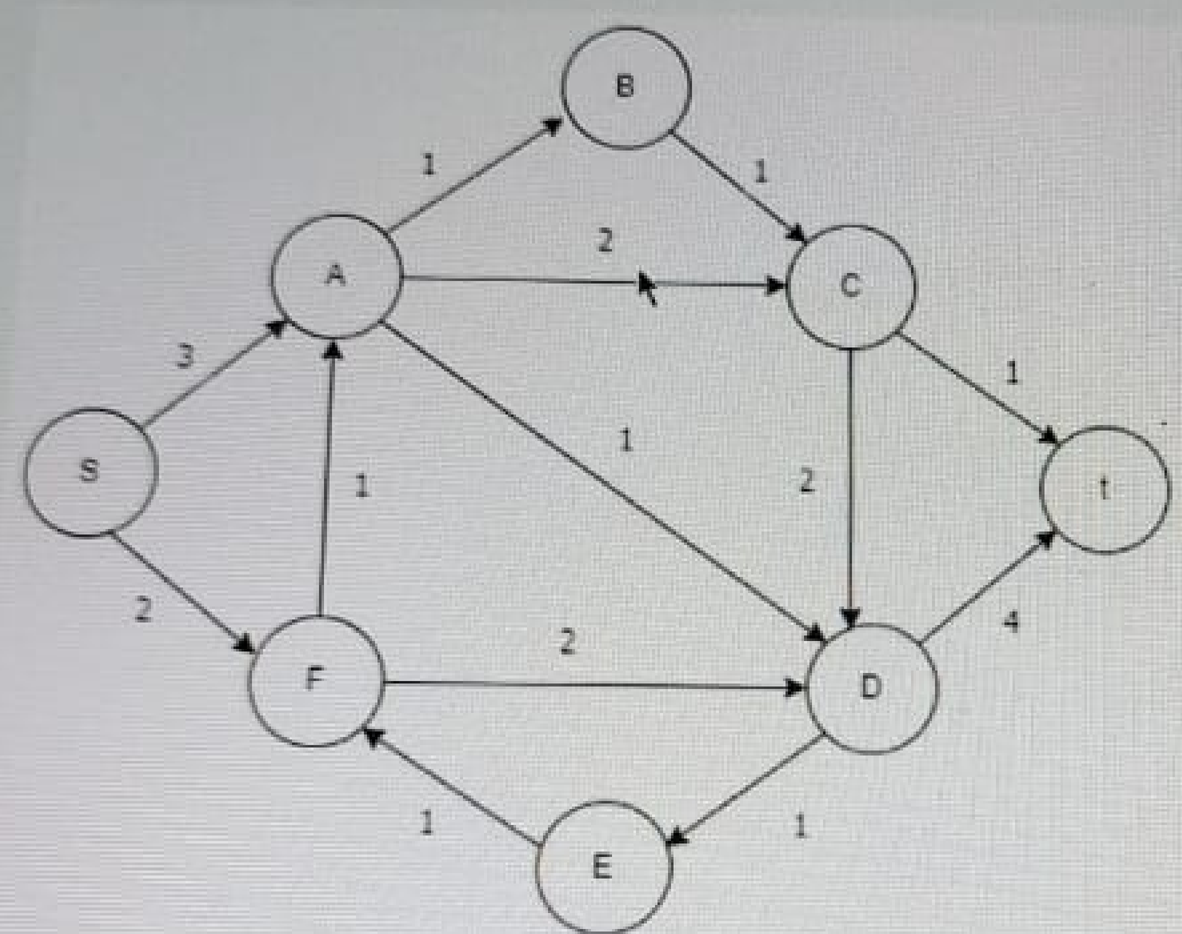
12. (Programming Fundamentals) Flow Cut



What is the minimum s-t cut capacity of the above flow network?

Pick **ONE** option

- 4
- 5

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13. (C Question) enum Question

Based on the above code, what will be the values of Aug and Nov respectively?

```
#include <stdio.h>

int main()

{
                               9    10
enum year{Jan, Feb, Mar=8, Apr, May, Jun, Jul,
        13         6    7    8
        Aug, Sep=5, Oct, Nov, Dec};

  return 0;

}
```

Pick **ONE** option

- 4 and 7
- 13 and 7 ✓

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```
int main()
{
enum year{Jan, Feb, Mar=8, Apr, May, Jun, Jul,
        Aug, Sep=5, Oct, Nov, Dec};
    return 0;
}
```

Pick **ONE** option

- 4 and 7
- 13 and 7
- 8 and 11
- 13 and 16

Clear Selection

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14. (C Question) Loop

How many times 'MATLAB' will be printed in the following C code?

```
#include <stdio.h>
int main()
{
    int i = 1024;
    for (; i; i >>= 1)
        printf("MATLAB");
    return 0;
}
```

11

Pick **ONE** option

- 10

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17. (C Question) Pointer arithmetic

What is the output of the following code snippet?

```
#include <stdio.h>
void main()
{
int x = 4;
int *p = &x;
int *k = p++;
int r = p - k;
printf("%d", r);
}
```

Pick **ONE** option

- 4
- 2
- 1
- 5

Clear Selection

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18. (C Question) Correct the code snippet

How can the issue in the following code be solved? Select all that apply.

```
int* add_numbers(int, int);
void main()
{
int* p;
p = add_numbers(1, 3);
}
int* add_numbers(int a, int b)
{
int* sum = (int*) malloc (16);
*sum = a + b;
return sum;
}
```

NO ISSUE

Pick **ONE OR MORE** options

- [x] Defining the function add_numbers above main function
- [x] By making the variable declaration inside the function as static
- [x] By deallocating the allocated variable
- [x] By making the variable sum global

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19. (C Question) Pointers in C

While thinking of pointers, how many levels of indirection does ANSI C standard say you mu

Pick **ONE** option

- At least 12
- At most 12
- At least 3
- At most 3
- Exactly 3

Clear Selection

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20. (C Question) Pointers 1

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Predict the output of following program:

```
#include<stdio.h>
void main()
{
int a[]={1,2,4,6,8};
int* p[]={a,a+1,a+2,a+3,a+4};
int** p1=p;
int* p2=*(p+2);
printf("%u %u %u\n",*++p2,++*p2++,*++*++*++p1);
}
```

Pick **ONE** option

- 6 7 4
- 9 7 3

```
int* p[]={a,a+1,a+2,a+3,a+4};

int** p1=p;

int* p2=*(p+2);

printf("%u %u %u\n",***p2,***p2++,******p1);

}
```

Pick **ONE** option

- 6 7 4
- 9 7 3
- 9 8 3
- 8 5 4

Clear Selection

Continue

24. (C++ Question) String printer

What would be the output of the following code?

```
#include <iostream>

int main(){
        std::string s("mathworks");
        for(;s!='\0';s++)
            printf("%c", s);
}
```

Pick **ONE** option

- mathworks
- athworks
- Compiler error
- No output

Clear Selection

ALL

25. (C++) Find the output of the following question.

21 23 24 25 26 27

```
#include <iostream>
using namespace std;
int main()
{
    int a[5] = {1,2,3,4,5};
    int *ptr;
    ptr = a;
    cout << *++ptr << endl;
    cout << *(ptr++) << endl;
    cout << *ptr << endl;

    return 0;
}
```

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```
    return 0;
}
```

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Pick **ONE** option

- 1
  2
  3
- 1
  2
  2
- 2
  2
  2
- 2
  2
  3

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26. (C++ Question) VTable

When is a VTable created?

Pick **ONE** option

- Every Class has Virtual Table
- Class inherited from other Class
- When a Class Overrides the function of Base class
- None of the above

Clear Selection

A VTable (Virtual Table) is created in C++ for a class that contains virtual functions. A VTable is used to support dynamic (run-time) polymorphism, where the actual function to be called is determined at run time rather than compile time.

A VTable is created when:

A class has at least one virtual function or overrides a virtual function from a base class.
Each object of such a class contains a pointer to the VTable, known as the vptr (virtual pointer), which enables runtime lookup for the correct function to call.

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27. (C++ Question) unsigned char Output

What would the following code output?

```
#include<iostream>
int main()
{
 char c;
 c = 128;
 printf("%d %d\n", c, (unsigned char)c);
 return 0;
}
```

-128 128

Pick **ONE** option

- 128 128
- Error

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What is the output of the following?

```
#include <iostream>
using namespace std;

class X
{
public: X()
        { cout<<"X"; }
        ~X()
        { cout<<"~X"; }
};

class Y : public X
{
public: Y()
        { cout<<"Y"; }
        ~Y()
        { cout<<"~Y"; }
};

int main()
{
    Y obj;
    return 0;
}
```

Pick **ONE** option

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```
class Y : public X
{
public: Y()
        { cout<<"Y"; }
        ~Y()
        { cout<<"~Y"; }
};

int main()
{
    Y obj;
    return 0;
}
```

Pick **ONE** option

- YX~X~Y
- XY~Y~X
- X~XY~Y
- X~X~YY

Clear Selection

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